LOGO	**COST ESTIMATE SUMMARY**	Document No. F-TSG-025 Revision No. 00 Revision Date N/A

PROJECT TYPE:	☐ Electrical ☑ Mechanical	**DATE:**	2026-10-03
PROJECT DESCRIPTION:	Overhaul of Mak-Ban Unit 3 cooling water pump, including a shop-fabricated spare shaft and traded supplier parts.	**COST ESTIMATE (CE) NUMBER** **CE-2026-014**	
CLIENT NAME:	Philippine Geothermal Production Company – Mak-ban	**REVISION NO.** **REVISION DATE**	
CLIENT LOCATION:	Brgy. Limao, Calauan, Laguna	**MATERIAL:**	Cooling water pump, Unit 3
ATTENTION:	Plant Maintenance Superintendent	**QUANTITY:**	1
END USER:	PGPC Mak-Ban	**NO. OF DAYS:**	11

ITEM	DESCRIPTION	TOTAL COST
A.	MANPOWER COST	₱202,710.94
B.	TOOLS AND EQUIPMENTS	₱136,896.34
C.	MATERIALS AND CONSUMABLES	₱26,964.20
D.	PERSONAL PROTECTIVE EQUIPMENT	₱21,561.00
E.	MISCELLANEOUS	₱138,700.00
E.1	ACCOMMODATION	₱75,600.00
E.2	TRANSPORTATION	₱44,100.00
E.3	REQUIREMENTS	₱0.00
E.4	ADMIN COST	₱10,500.00
E.5	INSURANCES	₱8,500.00
E.6	THIRD PARTY SERVICES	₱0.00
	TOTAL AMOUNT:	**₱526,832.48**
	UNIT PRICE:	**₱526,832.48**

1. Any additional scope not stated in the original scope of work is not included in this cost estimate.

2. Lead time assumes the crew is not held by the plant.

Prepared By:

Andre Villanueva 2026-10-01 16:20

Mr. Andre Villanueva

Cost Estimator

Checked By:

Liza Bautista 2026-10-02 11:10

Mrs. Liza Bautista

Costing Lead

Noted By:

Helen Navarro 2026-10-03 15:45

Mrs. Helen Navarro

Department Head

Approved By:

Helen Navarro 2026-10-03 15:45

Mrs. Helen Navarro

Department Head

Reviewed By:

Liza Bautista 2026-10-02 11:10

Mrs. Liza Bautista

Costing Lead

Checked By:

Paolo Cruz 2026-10-03 16:00

Mr. Paolo Cruz

Sales Manager

Concurred by:

Carla Mendoza 2026-10-03 16:05

Mrs. Carla Mendoza

Sales Representative

As found

Mak-Ban Unit 3 cooling water pump is vibrating above the plant limit. The shaft sleeve is scored and the coupling guard was removed for inspection. The client asked for an onsite overhaul, one shop-fabricated spare shaft, and supplier parts brought in under trading.

As left

Pump reassembled and aligned, spare shaft delivered with the QA/QC record, and the traded parts turned over at the warehouse.

Work type: ONSITE

Work type: SHOPWORK

Work type: TRADING

A. ONSITE WORKS

A. Regular Day (Dayshift)

Item No.	Role / Position	QTY	Days	Rate/Day	OT Hrs	Subtotal
1	Supervisor	1	4	₱2,500.00	1	₱11,562.50
2	Safety Officer II	1	2	₱900.00	1	₱2,081.25
3	Rigger	2	2	₱1,300.00	1	₱6,012.50
4	Scaffolder	3	2	₱1,300.00	1	₱9,018.75
5	Trade Assistant	2	2	₱800.00	0	₱3,200.00
6	Mechanical Fitter	1	7	₱1,300.00	2	₱11,943.75
7	Millwright	2	9	₱1,500.00	2	₱35,437.50
8	Welder	1	7	₱900.00	2	₱8,268.75
9	Electrician	1	2	₱900.00	2	₱2,362.50

B. Regular Night (Nightshift)

Item No.	Role / Position	QTY	Days	Rate/Day	OT Hrs	Subtotal
1	Mechanical Fitter	1	7	₱1,300.00	2	₱13,138.13
2	Millwright	1	7	₱1,500.00	2	₱15,159.38
3	Trade Assistant	1	7	₱800.00	2	₱8,085.00

E. Holiday Day (Dayshift)

Item No.	Role / Position	QTY	Days	Rate/Day	OT Hrs	Subtotal
1	Mechanical Fitter	1	7	₱1,300.00	2	₱23,887.50
2	Millwright	1	7	₱1,500.00	2	₱27,562.50
3	Welder	1	7	₱900.00	2	₱16,537.50

F. Holiday Night (Nightshift)

Item No.	Role / Position	QTY	Days	Rate/Day	OT Hrs	Subtotal
1	Mechanical Fitter	1	7	₱1,300.00	2	₱26,276.25
2	Millwright	1	7	₱1,500.00	2	₱30,318.75
3	Trade Assistant	1	7	₱800.00	2	₱16,170.00

Subtotal — ONSITE	**₱267,022.50**

B. SHOP WORKS

A. Regular Day (Dayshift)

Item No.	Role / Position	QTY	Days	Rate/Day	OT Hrs	Subtotal
1	Machinist	1	5	₱900.00	2	₱5,906.25
2	Welder	1	5	₱900.00	2	₱5,906.25
3	Trade Assistant	1	5	₱800.00	0	₱4,000.00
4	Qa/qc	1	2	₱1,500.00	1	₱3,468.75
5	NDT Technician	1	2	₱1,300.00	1	₱3,006.25
6	Driver	1	1	₱750.00	2	₱984.38
7	Rigger	1	1	₱1,300.00	1	₱1,503.13

Subtotal — SHOP	**₱24,775.00**

C. TRADING

A. Regular Day (Dayshift)

Item No.	Role / Position	QTY	Days	Rate/Day	OT Hrs	Subtotal
1	Driver	1	2	₱750.00	1	₱1,734.38
2	Trade Assistant	1	1	₱800.00	0	₱800.00
3	Qa/qc	1	1	₱1,500.00	1	₱1,734.38
4	Rigger	2	1	₱1,300.00	1	₱3,006.25
				Subtotal — TRADING		**₱7,275.00**

GRAND TOTAL	**₱299,072.50**

LOGO	**BILL OF TOOLS & EQUIPMENT**	Document No. F-TSG-025 Revision No. 00 Revision Date N/A

A. ONSITE WORKS

Item No.	Item Description	Qty	Days	Rate / Unit Cost	kW	Hrs/Day	₱/kWh	Electricity	Subtotal
1	Chain Block 2T	1	2	₱772.91	—	—	—	—	₱1,545.82
2	Chain Block 1T	2	2	₱227.08	—	—	—	—	₱908.32
3	Torque Wrench 1/2 In Drive	2	7	₱892.80	—	—	—	—	₱12,499.20
4	Angle Grinder 7 In	2	7	₱594.00	—	—	—	—	₱8,316.00
5	Alignment Tools Exworks Joliet	1	9	₱8,151.00	—	—	—	—	₱73,359.00
								Subtotal — ONSITE	**₱96,628.34**

B. SHOP WORKS

Item No.	Item Description	Qty	Days	Rate / Unit Cost	kW	Hrs/Day	₱/kWh	Electricity	Subtotal
1	Lathe Machine Ca6150 X 2000mm	1	5	₱5,000.00	7.5	8	₱12.00	₱3,600.00	₱28,600.00
2	Portable Welding Machine	1	5	₱500.00	5	6	₱12.00	₱1,800.00	₱4,300.00
3	Angle Grinder 4 In	1	5	₱396.00	0.8	3	₱12.00	₱144.00	₱2,124.00
4	Forklift	1	1	₱2,622.00	—	—	—	₱0.00	₱2,622.00
	of which electricity								₱5,544.00
								Subtotal — SHOP	**₱37,646.00**

C. TRADING

Item No.	Item Description	Qty	Days	Rate / Unit Cost	kW	Hrs/Day	₱/kWh	Electricity	Subtotal
1	Forklift	1	1	₱2,622.00	—	—	—	₱0.00	₱2,622.00
								Subtotal — TRADING	**₱2,622.00**

GRAND TOTAL	**₱136,896.34**

A. ONSITE WORKS				
Item No.	**Item Description**	**Qty**	**Rate / Unit Cost**	**Subtotal**
1	Grinding Disc 7 In	70	₱380.00	₱26,600.00
2	Penetrant 500ml	14	₱464.80	₱6,507.20
3	Loctite 243	7	₱1,430.00	₱10,010.00
4	Gasket Sealant Den Braven 302ml	14	₱620.00	₱8,680.00
5	Rags Round White	35	₱65.00	₱2,275.00
6	Gadus Grease S2 V-100-3 (rl3)	2	₱9,945.00	₱19,890.00
			Subtotal — ONSITE	**₱73,962.20**

B. SHOP WORKS				
Item No.	**Item Description**	**Qty**	**Rate / Unit Cost**	**Subtotal**
1	Tungsten Electrode 2.4mm X 175mm	30	₱94.00	₱2,820.00
2	Cutting Disc 4 In	40	₱35.00	₱1,400.00
3	Grinding Disc 4 In	30	₱72.00	₱2,160.00
4	Penetrant 500ml	2	₱464.80	₱929.60
			Subtotal — SHOP	**₱7,309.60**

C. TRADING				
Item No.	**Item Description**	**Qty**	**Rate / Unit Cost**	**Subtotal**
1	O-ring Seal 90mm ID X 100mm OD X 5mm	4	₱800.00	₱3,200.00
2	Air Coupling 2 Legs 1 In	2	₱488.40	₱976.80
3	Loctite 586 Thread Sealant Henkel 50ml	1	₱3,377.00	₱3,377.00
			Subtotal — TRADING	**₱7,553.80**

GRAND TOTAL	**₱88,825.60**

A. ONSITE WORKS				
Item No.	**Item Description**	**Qty**	**Rate / Unit Cost**	**Subtotal**
1	Safety Helmet With Chin Strap	50	₱245.00	₱12,250.00
2	Safety Shoes	50	₱970.00	₱48,500.00
3	High Visibility Vest	8	₱145.00	₱1,160.00
4	Caution Tape	4	₱750.00	₱3,000.00
5	Full Body Harness With Lanyard And Shock Absorber	10	₱115.00	₱1,150.00
6	Scaffolding Gloves	10	₱25.00	₱250.00
7	Cotton Hand Gloves	84	₱16.25	₱1,365.00
8	Safety Clear Goggles	50	₱173.00	₱8,650.00
9	Welding Gloves	14	₱190.00	₱2,660.00
10	Welding Mask	7	₱280.00	₱1,960.00
			Subtotal — ONSITE	**₱80,945.00**

B. SHOP WORKS				
Item No.	**Item Description**	**Qty**	**Rate / Unit Cost**	**Subtotal**
1	Welding Gloves	5	₱190.00	₱950.00
2	Welding Mask	5	₱280.00	₱1,400.00
3	Safety Clear Goggles	10	₱173.00	₱1,730.00
4	Nitrile Hand Gloves	4	₱10.00	₱40.00
5	High Visibility Vest	2	₱145.00	₱290.00
			Subtotal — SHOP	**₱4,410.00**

C. TRADING				
Item No.	**Item Description**	**Qty**	**Rate / Unit Cost**	**Subtotal**
1	Nitrile Hand Gloves	1	₱10.00	₱10.00
2	Safety Shoes	3	₱970.00	₱2,910.00
			Subtotal — TRADING	**₱2,920.00**

GRAND TOTAL	**₱88,275.00**

A. ONSITE WORKS							
Work type	**Section**	**Item No.**	**Description**	**QTY**	**Days**	**Unit cost**	**Subtotal**
ONSITE	A	1	Hotel	6	7	₱1,800.00	₱75,600.00
ONSITE	B	2	Diesel	40	7	₱70.00	₱19,600.00
ONSITE	B	3	Vehicle Hilux/Pickup (PM)	1	7	₱3,500.00	₱24,500.00
ONSITE	D	4	Meal Allowance	6	7	₱250.00	₱10,500.00
ONSITE	I	5	Insurances	1	1	₱8,500.00	₱8,500.00
						Subtotal — ONSITE	**₱138,700.00**

GRAND TOTAL	**₱138,700.00**

Part A: Manpower Loading (11 schedule days)

Item	Manpower Loading (Role)	Qty	UOM	No. of Days	Rate Per Day	Sub-Total A	OT Hrs Per Day	Allowance	Sub-Total B	Total
1	Scaffolder	3	pax	11	₱1,300.00	₱42,900.00	1	₱200.00	₱6,600.00	₱49,500.00
2	Trade Assistant	4	pax	11	₱800.00	₱35,200.00	2	₱200.00	₱8,800.00	₱44,000.00
3	Supervisor	1	pax	11	₱2,500.00	₱27,500.00	1	₱200.00	₱2,200.00	₱29,700.00
4	Electrician	1	pax	11	₱900.00	₱9,900.00	2	₱200.00	₱2,200.00	₱12,100.00
5	Millwright	4	pax	11	₱1,500.00	₱66,000.00	2	₱200.00	₱8,800.00	₱74,800.00
6	Mechanical Fitter	4	pax	11	₱1,300.00	₱57,200.00	2	₱200.00	₱8,800.00	₱66,000.00
7	Welder	2	pax	11	₱900.00	₱19,800.00	2	₱200.00	₱4,400.00	₱24,200.00
8	Safety Officer II	1	pax	11	₱900.00	₱9,900.00	1	₱300.00	₱3,300.00	₱13,200.00
9	Rigger	2	pax	11	₱1,300.00	₱28,600.00	1	₱200.00	₱4,400.00	₱33,000.00
									Subtotal — Part A	**₱297,000.00**

Part B: Allowance

Item	Category	Qty	UOM	No. of Days	Rate	Total
2	Skilled	21	pax	11	₱200.00	₱46,200.00
3	Admin	1	pax	11	₱300.00	₱3,300.00
					Subtotal — Part B	**₱49,500.00**

Part C: Line Items

Line Item	Description/Manpower	Qty	UOM	No. of Days / Trips	Unit Cost	Total
					Subtotal — Part C	**₱0.00**
					TOTAL	**₱346,500.00**

Part A: Manpower Loading (11 schedule days)

Item	Manpower Loading (Role)	Qty	UOM	No. of Days	Rate Per Day	Sub-Total A	OT Hrs Per Day	Allowance	Sub-Total B	Total
1	Scaffolder	3	pax	11	₱1,300.00	₱42,900.00	1	₱200.00	₱6,600.00	₱49,500.00
2	Trade Assistant	4	pax	11	₱800.00	₱35,200.00	2	₱200.00	₱8,800.00	₱44,000.00
3	Supervisor	1	pax	11	₱2,500.00	₱27,500.00	1	₱200.00	₱2,200.00	₱29,700.00
4	Electrician	1	pax	11	₱900.00	₱9,900.00	2	₱200.00	₱2,200.00	₱12,100.00
5	Millwright	4	pax	11	₱1,500.00	₱66,000.00	2	₱200.00	₱8,800.00	₱74,800.00
6	Mechanical Fitter	4	pax	11	₱1,300.00	₱57,200.00	2	₱200.00	₱8,800.00	₱66,000.00
7	Welder	2	pax	11	₱900.00	₱19,800.00	2	₱200.00	₱4,400.00	₱24,200.00
8	Safety Officer II	1	pax	11	₱900.00	₱9,900.00	1	₱300.00	₱3,300.00	₱13,200.00
9	Rigger	2	pax	11	₱1,300.00	₱28,600.00	1	₱200.00	₱4,400.00	₱33,000.00
									Subtotal — Part A	**₱297,000.00**

Part B: Allowance

Item	Category	Qty	UOM	No. of Days	Rate	Total
2	Skilled	21	pax	11	₱200.00	₱46,200.00
3	Admin	1	pax	11	₱300.00	₱3,300.00
					Subtotal — Part B	**₱49,500.00**

Part C: Line Items

Line Item	Description/Manpower	Qty	UOM	No. of Days / Trips	Unit Cost	Total
					Subtotal — Part C	**₱0.00**
					TOTAL	**₱346,500.00**

Part A: Manpower Loading (8 schedule days)

Item	Manpower Loading (Role)	Qty	UOM	No. of Days	Rate Per Day	Sub-Total A	OT Hrs Per Day	Allowance	Sub-Total B	Total
1	Driver	2	pax	8	₱750.00	₱12,000.00	2	₱200.00	₱3,200.00	₱15,200.00
2	Rigger	3	pax	8	₱1,300.00	₱31,200.00	1	₱200.00	₱4,800.00	₱36,000.00
3	Qa/qc	2	pax	8	₱1,500.00	₱24,000.00	1	₱200.00	₱3,200.00	₱27,200.00
4	NDT Technician	1	pax	8	₱1,300.00	₱10,400.00	1	₱200.00	₱1,600.00	₱12,000.00
5	Machinist	1	pax	8	₱900.00	₱7,200.00	2	₱200.00	₱1,600.00	₱8,800.00
6	Welder	1	pax	8	₱900.00	₱7,200.00	2	₱200.00	₱1,600.00	₱8,800.00
7	Trade Assistant	2	pax	8	₱800.00	₱12,800.00	0	₱200.00	₱3,200.00	₱16,000.00
									Subtotal — Part A	**₱104,800.00**

Part B: Allowance

Item	Category	Qty	UOM	No. of Days	Rate	Total
2	Skilled	12	pax	8	₱200.00	₱19,200.00
					Subtotal — Part B	**₱19,200.00**

Part C: Line Items

Line Item	Description/Manpower	Qty	UOM	No. of Days / Trips	Unit Cost	Total
					Subtotal — Part C	**₱0.00**
					TOTAL	**₱124,000.00**

Part A: Manpower Loading (8 schedule days)

Item	Manpower Loading (Role)	Qty	UOM	No. of Days	Rate Per Day	Sub-Total A	OT Hrs Per Day	Allowance	Sub-Total B	Total
1	Driver	2	pax	8	₱750.00	₱12,000.00	2	₱200.00	₱3,200.00	₱15,200.00
2	Rigger	3	pax	8	₱1,300.00	₱31,200.00	1	₱200.00	₱4,800.00	₱36,000.00
3	Qa/qc	2	pax	8	₱1,500.00	₱24,000.00	1	₱200.00	₱3,200.00	₱27,200.00
4	NDT Technician	1	pax	8	₱1,300.00	₱10,400.00	1	₱200.00	₱1,600.00	₱12,000.00
5	Machinist	1	pax	8	₱900.00	₱7,200.00	2	₱200.00	₱1,600.00	₱8,800.00
6	Welder	1	pax	8	₱900.00	₱7,200.00	2	₱200.00	₱1,600.00	₱8,800.00
7	Trade Assistant	2	pax	8	₱800.00	₱12,800.00	0	₱200.00	₱3,200.00	₱16,000.00
									Subtotal — Part A	**₱104,800.00**

Part B: Allowance

Item	Category	Qty	UOM	No. of Days	Rate	Total
2	Skilled	12	pax	8	₱200.00	₱19,200.00
					Subtotal — Part B	**₱19,200.00**

Part C: Line Items

Line Item	Description/Manpower	Qty	UOM	No. of Days / Trips	Unit Cost	Total
					Subtotal — Part C	**₱0.00**
					TOTAL	**₱124,000.00**

LOGO

REQUEST FOR COSTING (RCE) CHECKLIST FORM

☑ NEW PROJECT ☐ EXISTING PROJECT	**INQUIRY NUMBER & DATE:** INQ-PGPC-2026-118 2026-09-26
CUSTOMER: Philippine Geothermal Production Company – Mak-ban	**RCE NO.:** RCE-2026-014
ADDRESS: Brgy. Limao, Calauan, Laguna	**RCE DATE:** 2026-09-28
PROJECT TITLE: Mak-Ban Unit 3 Cooling Water Pump Overhaul	**PRIORITY LEVEL:** High
PROJECT TYPE: ☑ SHOPWORK ☑ ON-SITE ☑ TRADING	**CE DEADLINE:** 2026-10-06
DEPARTMENT: ☑ MECHANICAL ☐ ELECTRICAL	**SUBMISSION DEADLINE:** 2026-10-10

Note: Please kindly attach files Needed.

ITEM NO.	DESCRIPTION	YES	NO	N/A	REMARKS
1	PR / ITB / RFQ / BUDGETARY INQUIRY (BI)	X			Inquiry INQ-PGPC-2026-118 received 26 Sep 2026.
2	IN-LINE WITH COMPANY 1 PRODUCTS AND SERVICES	X			Mechanical overhaul is in line with Company 1 services.
3	SITE INSPECTION / PRE-BID MEETING SCHEDULE & LOCATION	X			Site walkdown done 25 Sep 2026 at Mak-Ban Unit 3.
4	SITE CONDITIONS & LOGISTICS	X			Access by plant permit. Laydown beside the pump bay.
5	TERMS OF REFERENCE (TOR)	X			Terms of reference came with inquiry INQ-PGPC-2026-118.
6	DETAILED DESCRIPTION OF SCOPE OF WORKS	X			Onsite overhaul, shop spare shaft, traded seal parts.
8	ASSET AND EQUIPMENT DETAILS (DRAWING, PHOTOS, ETC.)	X			Pump GA and nameplate photo attached to the request.
9	MATERIAL SPECIFICATION	X			Shaft and seal sizes taken from the Unit 3 nameplate.
10	CONSUMABLES AND SPARE (IF ANY)	X			Traded seals and one shop-made spare shaft. Plant stock is not included.
11	ACCEPTANCE CRITERIA	X			Vibration within the plant limit and a leak-free run test.
12	COMPLETION LEAD TIME / WORK DURATION	X			11 days onsite window, shop and trading inside the same period.
13	IMPLEMENTATION SCHEDULE	X			Isolation and scaffold together, then the 7-day overhaul, then run test.
14	RECOMMENDATION	X			Accept and release to costing.
14.1	TO PROCEED THE RCE	X			Proceed.
14.2	TO SECURE COMPLETE REFERENCE DATA			X	Reference data is enough to quote.
14.3	TO DECLINE / NO QUOTE			X	Quote will be issued.

OTHER REMARKS:
Hybrid job. Onsite crew of three per shift, day and night, for seven days including one regular holiday with overtime. Shop fabricates one spare shaft. Trading covers pickup, incoming inspection, and delivery of supplier parts.

REASON TO DECLINE / NO QUOTE:

Prepared by:	Reviewed by:	Approved by:
Carla Mendoza Sales Representative	Paolo Cruz Sales Manager	Helen Navarro Department Head
Carla Mendoza 2026-09-28 08:15	Paolo Cruz 2026-09-28 14:40	Helen Navarro 2026-09-29 09:05